

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Reveal Answer

1. What is the key architectural difference between STDIO and Streamable HTTP transports in MCP?

1 / 1 point

- ☐ STDIO uses a bidirectional HTTP stream while Streamable HTTP uses stdin and stdout
- ☒ STDIO is process-local, while Streamable HTTP supports remote network communication
- ☐ Both transports require separate request and response channels
- ☐ Streamable HTTP requires no authentication, while STDIO always requires authentication

Nice work

STDIO runs as a local subprocess, while Streamable HTTP enables remote, network-based MCP servers.

2. What role does sampling play in MCP?

2 / 1 point

- ☐ It is always enabled in every MCP client
- ☐ It automatically runs LLMs inside the server without user involvement
- ☐ It replaces JSON-RPC messaging between client and server
- ☒ It allows a server tool to request LLM generation through the client when additional reasoning is needed

Nice work

Server-side sampling:toolMessage request, the client obtains approval, calls the LLM, and returns the response.

3. What is the primary security purpose of Roots in MCP?

3 / 1 point

- ☐ They encrypt all files accessed by MCP servers
- ☐ They define which users are allowed to start servers
- ☐ They ensure HTTP authentication for remote MCP servers
- ☒ They establish allowed filesystem directories and prevent access outside declared paths

Nice work

Roots define security boundaries, validated through canonical path resolution to block traversal or unauthorized access.

4. Why does the server validate file paths using a helper function such as is_within_roots before reading or writing files?

4 / 1 point

- ☐ To automatically format file contents before returning them
- ☐ To detect syntax errors in file contents before execution
- ☒ To ensure that all requested file operations stay within the approved workspace directory
- ☐ To optimize file access performance by caching directories

Nice work

Path validation prevents path traversal and unauthorized filesystem access.

5. Why does an MCP server expose file operations such as reading, writing, and listing files as callable tools rather than embedding them directly into the server runtime?

5 / 1 point

- ☒ To allow AI agents and clients to invoke file operations in a controlled, capability-based manner
- ☐ So that tools can execute only during server startup
- ☐ Because tools automatically bypass security constraints
- ☐ To eliminate the need for input validation in file operations

Nice work

Tools define clear, structured entry points that MCP clients can call securely.